Controlling – General Data

1. General

The propeller is controlled by the propeller and engine power levers in conjunction with a constant speed governor, overspeed governor, and autofeather system. An overspeed governor test facility is provided for testing the overspeed governor on the ground. When aircraft operated on floats, the blade zero thrust stop system (S.O.O. 6203) needs to be fitted. A propeller tachometer indicating system and a Reset Props caution message is provided through the CAS.

2. Description

The constant speed governor combines the functions of a normal constant speed unit (CSU), beta range control valve, and fuel governor. As a CSU, it regulates power turbine speed by varying the pitch of the propeller to match the load torque to engine torque in response to varying conditions of flight. The beta range control valve controls the propeller blade angles in the beta range as selected by the engine power lever. The beta range is that segment in the overall pitch range of the propeller, which is directly controllable by the power lever; it extends from +17 degrees in the forward range to –15 degrees in the reverse range. During beta range operation, the propeller is kept in the under speeding condition as a function of airspeed. The fuel governor reset lever, which is linked to the power lever control operating the reversing lever, moves progressively to reset a lower maximum power turbine speed, relevant to the selected speed, to limit the power output of the gas generator, by restricting fuel flow to a value which prevents propeller speed exceeding a value 5% below selected propeller speed. The fuel governor section of the constant speed governor also protects the engine against propeller overspeeding, in the event of an overspeed governor failure during normal forward thrust operation, by reducing fuel flow.

The overspeed governor provides automatic control of a propeller overspeed condition by increasing the pitch of the propeller blades to absorb engine power, thereby decreasing propeller speed. The overspeed test switch, when operated on the ground, simulates an overspeed condition of the propeller by resetting the governor to a lower value.

Propeller feathering is accomplished by selecting the propeller lever to FEATHER.

A propeller lever/power lever interlock mechanism is fitted, which prevents movement of the power levers below IDLE, if both propeller levers are positioned at less than 91% propeller rpm. Individual operation of either propeller lever above 91% rpm disengages the interlock, to permit the power levers to be moved below IDLE.

3. Operation

A. Propeller On-speed

In the on-speed condition, forces acting on the engine, propeller and propeller constant speed governor are in balance. The pilot valve within the CSU section of the governor has closed off the oil ports to meter only sufficient oil to the propeller servo piston, to maintain blade pitch.

B. Propeller Overspeed

In an overspeed condition, the governor mechanism causes the pilot valve to lift and uncover oil ports so that oil acting on the propeller servo piston is allowed to return to the reduction gear sump. This results in the propeller piston moving rearwards under the combined influence of the blade counterweights and feather return springs, to turn the blades into a higher pitch. Increasing blade pitch decreases propeller speed and consequently governor speed, so that the pilot valve moves to close off the oil ports when an on-speed condition is reached. In the event of a propeller maximum overspeed condition occurring, the propeller overspeed governor mechanism lifts the overspeed governor pilot valve, to allow propeller servo piston pressure oil to be damped to the reduction gearbox, so that the blades can move to increase their pitch and so reduce propeller speed.

C. Propeller Underspeed

In an underspeed condition, the constant speed governor mechanism lowers the pilot valve and uncovers ports to allow pressure oil to be directed to the propeller servo piston, which moves forward and turns the blades into a lower pitch, so allowing propeller speed to increase till the on-speed condition is reached.

D. Reverse Thrust Operation

During reverse thrust operation when the power lever is moved into the REVERSE range, the reversing lever and consequently the beta control valve is moved rearwards. In this position the beta control valve allows pressure oil to act on the propeller servo piston to turn the blades into reverse pitch. As the propeller piston moves forward, the low stop collar also moves forward and operates the beta control valve to close off the oil supply, so that when the propeller blades reach the pitch determined by power lever selection, the oil supply to the servo piston is just sufficient to maintain a balance.

E. Propeller Feathering

When the propeller lever is selected to feather, the CSU pilot valve is lifted and oil pressure acting on the propeller servo piston is relieved. This permits the combined effect of the blade counterweights and the feather return springs, to turn the blades into the feathered position (a pitch of +87 degrees).

F. Propeller Unfeathering

To unfeather a propeller, the propeller lever must be moved to full INCREASE to reset the constant speed governor, which in effect moves the pilot valve to open the governor oil ports, so that when the engine is started, pressure oil from the governor pump is directed to the propeller servo piston to turn the blades into low pitch. As the propeller piston moves forward, the low stop collar also moves forward and operates the beta control valve to close off the oil supply, so that when the blades reach the pitch determined by power lever selection, the oil supply to the servo piston is sufficient to maintain a balance.